

[INTERNAL]

Redis Caching Design — API Gateway & Semantic Cache Layer

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Last Updated: 2026-02-14

Classification: internal

This document specifies the Redis caching strategy for NovaTech's API Gateway and the new semantic cache layer introduced in Project Atlas v2.1. The API Gateway caches authentication tokens in Redis with a TTL of 15 minutes; internal benchmarks showed a 42% reduction in PostgreSQL read load after rollout. A semantic cache, keyed on embedding similarity, cuts LLM inference costs by approximately 31% for repeated or near-duplicate queries.

1. Redis Topology

Production runs a 3-node Redis Cluster (ElastiCache r7g.large) with 1 primary and 1 replica per shard, spread across 3 AZs. Cluster mode is enabled with 3 shards, providing 18 GB total memory and automatic failover within 60 seconds.

Cache Namespace	TTL	Eviction Policy	Est. Key Count
auth:tokens	15 min	volatile-lru	40 000
embed:cache	24 h	volatile-lru	500 000
sem:cache	6 h	volatile-lfu	80 000
rate:limit	1 min	volatile-ttl	120 000
session:ctx	30 min	volatile-lru	25 000

2. Semantic Cache Design

The semantic cache stores (query_embedding → {response, source_chunks}) pairs. On every /ask request, the Retriever computes the query embedding and performs a nearest-neighbour lookup in a dedicated Qdrant collection (*sem_cache*) with a cosine similarity threshold of 0.96. A hit bypasses the full RAG pipeline and returns the cached response in <15 ms vs the normal 210 ms median.

Cache write-back uses a Celery background task to avoid blocking the request path. The task serialises the response, source chunk IDs, and timestamp to Redis using MessagePack encoding (saves ~35% vs JSON for our payload shapes).

3. Token Cache Deep Dive

Keycloak-issued JWTs are validated at the AuthMiddleware and stored in Redis as HASH structures: key = auth:token:{sha256(token)}, fields = user_id, roles, tenant_id, exp. This avoids repeated RS256 signature verification and Keycloak introspection calls. Benchmark: 0.8 ms (Redis hit) vs 38 ms (Keycloak round-trip).

- Cache is invalidated on token revocation via a pub/sub channel: auth:revoke.
- Revocation events are published by the Auth Service when a user logs out or is suspended.
- Cold start: first request after deployment always hits Keycloak; warm-up completes in ~30s.

4. Monitoring & Alerting

Cache hit rates are published as Prometheus metrics (redis_hit_ratio) and displayed on the NovaTech Ops Grafana dashboard. Alerting rules: hit ratio < 0.70 for auth:tokens over 5 minutes triggers a PagerDuty P2. Eviction rate > 5 000/s triggers a capacity review.